Elementary positivity for vertex multicolorings and dual Dyck functions

Abstract

We deduce coefficientwise elementary positivity for vertex multicolorings of natural unit interval graphs from the intermediate monomial-positivity results in OpenAI's chromatic-positivity preprint. A vertex of multiplicity a_i receives a set of a_i distinct colors. Extracting arbitrary nonnegative vertex exponents, rather than only squarefree exponents, extends the chromatic argument to these multiplicities without division by quantum factorials. This yields positivity for generalized Hessenberg chromatic functions. In a second part, a direct correspondence between step-τ dual Dyck factorizations and multicolorings proves elementary positivity in each statistic degree for every integer $\tau \geq 0$. An explicit grading shift recovers Hawkes's original dual Dyck symmetric functions at step one. All positivity conclusions use the stated external input.

1 Introduction

Elementary positivity asks for a symmetric function to be a nonnegative linear combination of products of elementary symmetric functions. For chromatic functions refined by an ascent statistic, the coefficientwise question requires the elementary coefficients to be polynomials with nonnegative integer coefficients. We consider the multicoloring version: each vertex receives a prescribed number of distinct colors, and adjacent vertices receive disjoint color sets. This construction appears in Hwang [Hwa] and Kiem–Lee [KL]; the multiplicities specify color-set cardinalities rather than powers of a single color.

The first part proves elementary positivity for every multiplicity vector on a natural unit interval graph (Theorem 3.1). The central observation is that the monomial-positivity results of [OAI] apply to all nonnegative vertex exponents. Restriction from a triangular graph and the elementary–monomial Cauchy identity then give the weighted theorem. This is a deduction from the specified intermediate results of [OAI], not an independent proof of chromatic positivity. The generalized Hessenberg functions of Kiem–Lee are one application, giving an affirmative answer to their Question 4.9 under the same input (Corollary 3.5).

The second part extends the dual Dyck factor conditions of Hawkes [Haw] to every integer step $\tau \geq 0$. These factorizations encode multicolorings of the graph joining distinct integer values at distance at most τ. Repeated values become vertex multiplicities, and the statistic is a constant shift of the complement of the weighted ascent statistic. Theorem 4.2 gives elementary positivity in each statistic degree. At step one, removing the equal-pair shift recovers Hawkes's original functions (Corollary 4.3), strengthening their Schur positivity in [Haw, Corollary 3.19].

The multiplicity step cannot be obtained simply by dividing an unweighted positive expansion. Replacing each vertex by a clique of its multiplicity introduces a factor $\prod_i [a_i]_q!$, where $[a]_q! = \prod_{j=1}^{a}(1 + q + \cdots + q^{j-1})$; see [KL, Remark 4.4]. Division by such a factor need not preserve coefficientwise positivity, as $(1+q)(1-q+q^2) = 1+q^3$ illustrates. The coefficient extraction below counts color sets directly.

2 Definitions and conventions

Let $\Lambda_{\mathbb{Z}} = \mathbb{Z}[e_1, e_2, \ldots]$ be the ring of symmetric functions in $\mathbf{x} = (x_1, x_2, \ldots)$, where $e_k(\mathbf{x}) = \sum_{c_1 < \cdots < c_k} x_{c_1} \cdots x_{c_k}$ and $e_0 = 1$. For a partition λ, write $|\lambda|$ for its size, $\ell(\lambda)$ for its length,

$e_\lambda = \prod_j e_{\lambda_j}$, and m_λ for the sum of distinct monomials with positive exponent multiset λ. A homogeneous symmetric function is *e-positive over* $\mathbb{Z}_{\geq 0}[q]$ when its e_λ-coefficients lie in $\mathbb{Z}_{\geq 0}[q]$.

An ordered graph G on $[r] = \{1, \ldots, r\}$ is a *natural unit interval graph* if, for a weakly increasing map $g : [r] \to [r]$ with $g(i) \geq i$, its edges are $\{i, j\}$ with $i < j \leq g(i)$. For $\mathbf{a} \in \mathbb{Z}^r_{\geq 0}$, set

$$N = |\mathbf{a}| := \sum_i a_i, \qquad M = M_G(\mathbf{a}) := \sum_{\substack{i<j \\ \{i,j\}\in E(G)}} a_i a_j. \tag{2.1}$$

A *proper multicoloring of multiplicity* $\mathbf{a}$ assigns a finite set $C_i \subseteq \mathbb{Z}_{>0}$ to each vertex, with $|C_i| = a_i$ and $C_i \cap C_j = \varnothing$ on every edge. Define

$$\begin{aligned} \mathrm{asc}_G(C) &= \sum_{\substack{i<j \\ \{i,j\}\in E(G)}} \#\{(c,d) \in C_i \times C_j : c < d\}, \\ X_{G,\mathbf{a}}(\mathbf{x}; q) &= \sum_{C \text{ proper}} q^{\mathrm{asc}_G(C)} \prod_{i=1}^{r} \prod_{c \in C_i} x_c. \end{aligned} \tag{2.2}$$

Put $R = \mathbb{Z}[v, v^{-1}]$. The *nonnegative-exponent graph algebra* is $\mathcal{A}^+_G = \bigoplus_{\mathbf{b}\in\mathbb{Z}^r_{\geq 0}} RZ^{\mathbf{b}}$, with multiplication

$$Z^{\mathbf{b}} Z^{\mathbf{c}} = v^{\Omega_G(\mathbf{b},\mathbf{c})} Z^{\mathbf{b}+\mathbf{c}}, \qquad \Omega_G(\mathbf{b}, \mathbf{c}) = \sum_{\substack{i<j \\ \{i,j\}\in E(G)}} (b_j c_i - b_i c_j). \tag{2.3}$$

Bilinearity of Ω_G gives associativity. The $Z^{\mathbf{b}}$ are basis symbols, and $[Z^{\mathbf{b}}]$ denotes coefficient extraction in this basis. For an independent set J, let $\mathbf{1}_J$ be its indicator vector and put

$$E_k(G) = \sum_{\substack{J\subseteq[r] \text{ independent} \\ |J|=k}} Z^{\mathbf{1}_J}, \qquad E_0(G) = 1. \tag{2.4}$$

By [OAI, Proposition 2.2], these elements commute. Thus $\Phi_G : \Lambda_{\mathbb{Z}} \to \mathcal{A}^+_G$ is the homomorphism determined by $\Phi_G(e_k) = E_k(G)$. Its degree is the total vertex exponent. All graph notation also applies to any finite totally ordered vertex set by increasing relabeling.

3 Elementary positivity for vertex multicolorings

External dependency. We take Proposition 2.2, Lemmas 5.1 and 5.7, and Theorem 5.8 of the September 24, 2026 version of [OAI] as external results. The proof below is a deduction from those assertions; it does not independently establish their validity.

Theorem 3.1. *Assuming the cited external results, for every natural unit interval graph G and every $\mathbf{a} \in \mathbb{Z}^{V(G)}_{\geq 0}$,*

$$X_{G,\mathbf{a}}(\mathbf{x}; q) = \sum_{\lambda \vdash N} c_\lambda(G, \mathbf{a}; q) e_\lambda(\mathbf{x}), \qquad c_\lambda(G, \mathbf{a}; q) \in \mathbb{Z}_{\geq 0}[q]. \tag{3.1}$$

More precisely, with M as in (2.1)*,*

$$c_\lambda(G, \mathbf{a}; q) = \sum_{d=0}^{M} [v^{\,2d-M} Z^{\mathbf{a}}] \Phi_G(m_\lambda)\, q^d. \tag{3.2}$$

In particular, each $c_\lambda(G, \mathbf{a}; q)$ has degree at most M.

3.1 Positivity at arbitrary vertex multiplicities

Lemma 3.2. *Under the external results above, for every natural unit interval graph G, every partition λ, and every $\mathbf{a} \in \mathbb{Z}_{\geq 0}^{V(G)}$,*

$$[Z^{\mathbf{a}}]\Phi_G(m_\lambda) \in \mathbb{Z}_{\geq 0}[v, v^{-1}]. \tag{3.3}$$

Proof. Choose an ordered induced embedding of G into an auxiliary triangular graph Γ supplied by [OAI, Lemma 5.1]. Write D for its independence number. The two results [OAI, Lemma 5.7 and Theorem 5.8], applied in the all-negative chamber of that construction, give

$$\Phi_\Gamma(m_\lambda) \in \bigoplus_{\mathbf{b} \in \mathbb{Z}_{\geq 0}^{V(\Gamma)}} \mathbb{Z}_{\geq 0}[v, v^{-1}]Z^{\mathbf{b}} \qquad \text{if } \ell(\lambda) \leq D. \tag{3.4}$$

Specifically, Theorem 5.8 identifies this evaluation with the section whose finiteness, nonnegative vertex support, and Laurent-coefficient positivity are asserted in Lemma 5.7. Thus (3.4) concerns every vertex exponent, not only exponents with coordinates zero or one.

To compare their finite-alphabet convention with ours, note that $E_k(\Gamma) = 0$ for $k > D$. Hence Φ_Γ factors through the specialization of symmetric functions to D variables. This agrees with the evaluation in the cited results. If $\ell(\lambda) > D$, that specialization sends m_λ to zero, so (3.4) holds for these partitions as well.

Identify $V(G)$ with the selected vertices of Γ and define the R-linear map

$$\rho : \mathcal{A}_\Gamma^+ \longrightarrow \mathcal{A}_G^+, \qquad \rho(Z^{\mathbf{b}}) = \begin{cases} Z^{\mathbf{b}|_{V(G)}}, & \operatorname{supp}(\mathbf{b}) \subseteq V(G), \\ 0, & \text{otherwise}, \end{cases} \tag{3.5}$$

where $\operatorname{supp}(\mathbf{b}) = \{u : b_u \neq 0\}$. This is an algebra homomorphism. Indeed, if either of two nonnegative exponent vectors uses an outside vertex, their sum still uses that vertex; both sides of multiplicativity are then zero. If both are supported on $V(G)$, the induced embedding gives $\Omega_\Gamma(\mathbf{b}, \mathbf{c}) = \Omega_G(\mathbf{b}|_{V(G)}, \mathbf{c}|_{V(G)})$, which proves multiplicativity in the remaining case.

An independent set of Γ contained in $V(G)$ is exactly an independent set of G. Therefore $\rho(E_k(\Gamma)) = E_k(G)$ for every k, and equality on the algebraically independent generators e_k implies

$$\rho \circ \Phi_\Gamma = \Phi_G. \tag{3.6}$$

Let $\widetilde{\mathbf{a}}$ extend $\mathbf{a}$ by zero outside $V(G)$. Then

$$[Z^{\mathbf{a}}]\Phi_G(m_\lambda) = [Z^{\widetilde{\mathbf{a}}}]\Phi_\Gamma(m_\lambda) \in \mathbb{Z}_{\geq 0}[v, v^{-1}],$$

as required. Crucially, (3.5) is defined on the nonnegative-exponent algebra, not on the full quantum torus, where vertex generators would be invertible. □

3.2 The multicoloring kernel and its normalization

For $s \geq 1$, write $\mathbf{x}_s = (x_1, \ldots, x_s)$ and set

$$\mathcal{E}_G(t) = \sum_{k \geq 0} E_k(G)t^k, \qquad K_s = \mathcal{E}_G(x_s)\mathcal{E}_G(x_{s-1}) \cdots \mathcal{E}_G(x_1). \tag{3.7}$$

The sum defining $\mathcal{E}_G$ is finite. Its values commute, but the displayed descending color order will match the ascent convention.

Lemma 3.3. *For every $s \geq 1$,*

$$X_{G,\mathbf{a}}(\mathbf{x}_s; v^2) = v^M [Z^{\mathbf{a}}] K_s. \tag{3.8}$$

Proof. Expanding K_s chooses an independent set J_c for each color $c \in [s]$. The associated term is

$$\left(\prod_{c=1}^{s} x_c^{|J_c|} \right) Z^{\mathbf{1}_{J_s}} Z^{\mathbf{1}_{J_{s-1}}} \cdots Z^{\mathbf{1}_{J_1}}.$$

Its vertex exponent is $\sum_c \mathbf{1}_{J_c}$. Thus extraction of $Z^{\mathbf{a}}$ is equivalent to requiring $C_i := \{c \in [s] : i \in J_c\}$ to have cardinality a_i. Independence of each J_c says exactly that adjacent vertices receive disjoint color sets. Conversely, the sets C_i recover all the J_c uniquely. This is a bijection with the multicolorings in (2.2); no occurrences at a vertex are labeled.

Repeated use of (2.3) gives the Laurent weight $v^{T(C)}$, where

$$T(C) = \sum_{c>d} \Omega_G(\mathbf{1}_{J_c}, \mathbf{1}_{J_d}). \tag{3.9}$$

Fix an edge $\{i, j\}$ with $i < j$ and a pair $(a, b) \in C_i \times C_j$. The colors are unequal. If $a < b$, the factor containing vertex j occurs to the left of the factor containing i, and its contribution to (3.9) is $+1$. If $a > b$, the contribution is -1. There are $a_i a_j$ such pairs on this edge. Consequently,

$$T(C) = \operatorname{asc}_G(C) - (M - \operatorname{asc}_G(C)) = 2 \operatorname{asc}_G(C) - M. \tag{3.10}$$

Multiplication by v^M changes the weight to $v^{2 \operatorname{asc}_G(C)}$, proving (3.8). □

3.3 Extraction of the elementary coefficients

Proof of Theorem 3.1. If $N = 0$, the unique multicoloring has all color sets empty and the function is 1; assume $N > 0$. In a second commuting alphabet $\mathbf{y}$, the elementary–monomial Cauchy identity is

$$\prod_{c=1}^{s} \left(\sum_{k \geq 0} e_k(\mathbf{y}) x_c^k \right) = \prod_{c,j} (1 + x_c y_j) = \sum_{\lambda} e_\lambda(\mathbf{x}_s) m_\lambda(\mathbf{y}). \tag{3.11}$$

For clarity, a fixed monomial in the y_j with positive exponents $\lambda_1, \ldots, \lambda_\ell$ has coefficient $\prod_{i=1}^{\ell} e_{\lambda_i}(\mathbf{x}_s)$ in the middle product. Grouping such monomials proves the last equality. The identity is interpreted degree by degree, so there is no convergence issue.

Apply Φ_G to the $\mathbf{y}$-symmetric functions. Since the $E_k(G)$ commute, their generating factors may be put in the descending order of (3.7), giving

$$K_s = \sum_{\lambda} e_\lambda(\mathbf{x}_s) \Phi_G(m_\lambda). \tag{3.12}$$

The evaluation preserves total degree. Extracting $Z^{\mathbf{a}}$ therefore retains only partitions of N. Combining (3.12) with Lemma 3.3, we obtain, for every s,

$$X_{G,\mathbf{a}}(\mathbf{x}_s; v^2) = \sum_{\lambda \vdash N} P_\lambda(v) e_\lambda(\mathbf{x}_s), \qquad P_\lambda(v) = v^M [Z^{\mathbf{a}}] \Phi_G(m_\lambda). \tag{3.13}$$

Lemma 3.2 gives $P_\lambda(v) \in \mathbb{Z}_{\geq 0}[v, v^{-1}]$. It remains to justify that these Laurent polynomials use only the even, nonnegative powers $1, v^2, \ldots, v^{2M}$; Laurent positivity alone would not establish the required conclusion over $\mathbb{Z}_{\geq 0}[q]$.

Take $s \geq N$. The polynomials $\{e_\lambda(\mathbf{x}_s) : \lambda \vdash N\}$ form a basis of the homogeneous symmetric polynomials of degree N. Every multicoloring satisfies $0 \leq \mathrm{asc}_G(C) \leq M$. Hence the left side of (3.13) has no power v^j with $j \notin \{0, 2, \ldots, 2M\}$. Extracting such a power gives

$$0 = \sum_{\lambda \vdash N} [v^j] P_\lambda(v)\, e_\lambda(\mathbf{x}_s).$$

Linear independence implies $[v^j]P_\lambda(v) = 0$ for every λ. Thus there is a polynomial $c_\lambda(q) \in \mathbb{Z}_{\geq 0}[q]$ of degree at most M such that $P_\lambda(v) = c_\lambda(v^2)$. Its coefficients are exactly those in (3.2). The substitution $q \mapsto v^2$ is injective, so (3.13) proves (3.1) in finitely many colors. These coefficients are independent of s, and setting $x_s = 0$ is compatible with the multicoloring definition. Passing to symmetric functions proves (3.1) in the infinite alphabet. □

Remark 3.4 (Why no factorial division is needed). At each vertex, the proof chooses a set of colors, not an ordering of those colors. Hence the coefficient $[Z^{\mathbf{a}}]$ already counts exactly the required multicolorings. The essential input is the all-exponent positivity of Lemma 3.2.

3.4 Generalized Hessenberg chromatic functions

Take $0 = i_0 < i_1 < \cdots < i_r = n$ and $I = \{i_1, \ldots, i_{r-1}\}$. A *generalized Hessenberg function* is a weakly increasing map $h : I \cup \{n\} \to I \cup \{n\}$ with $h(i) \geq i$. Its ordered weighted graph has vertices $I \cup \{n\}$, edges $\{i, j\}$ for $i < j \leq h(i)$, and weight $w_h(i_k) = i_k - i_{k-1}$. Kiem–Lee's function $\mathrm{csf}_q(h)$ is the multicoloring series (2.2) for this graph and weight [KL, Definitions 2.1 and 4.3].

Corollary 3.5. *Under the external input of Theorem 3.1,* $\mathrm{csf}_q(h)$ *is e-positive over* $\mathbb{Z}_{\geq 0}[q]$ *for every generalized Hessenberg function* h*.*

Proof. Define $g : [r] \to [r]$ by the requirement

$$h(i_k) = i_{g(k)}, \qquad a_k = i_k - i_{k-1}. \tag{3.14}$$

Since h is extensive and weakly increasing, $g(k) \geq k$ and g is weakly increasing. Increasing relabeling $i_k \mapsto k$ identifies the graph of h with the natural unit interval graph G_g: for $k < \ell$,

$$i_\ell \leq h(i_k) \quad \Longleftrightarrow \quad \ell \leq g(k).$$

The same relabeling preserves the color-set cardinalities and every ascending color pair across every edge. Accordingly,

$$\mathrm{csf}_q(h) = X_{G_g,\mathbf{a}}(\mathbf{x}; q), \qquad |\mathbf{a}| = \sum_{k=1}^{r} (i_k - i_{k-1}) = n. \tag{3.15}$$

Applying (3.1) to (3.15) yields

$$\mathrm{csf}_q(h) = \sum_{\lambda \vdash n} c_\lambda(G_g, \mathbf{a}; q) e_\lambda(\mathbf{x}), \qquad c_\lambda(G_g, \mathbf{a}; q) \in \mathbb{Z}_{\geq 0}[q].$$

This is precisely the elementary-positivity assertion of [KL, Question 4.9]. □

4 Elementary positivity for step-τ dual Dyck functions

Fix an integer $\tau \geq 0$. A *dual step-τ word* is a finite integer word w satisfying $w_{j+1} > w_j + \tau$ at each successive position. Empty and one-letter words are allowed. For a finite multiset S of integers, a dual step-τ factorization is a sequence $\mathcal{F} = (F_1, F_2, \ldots)$ of such words, all but finitely many empty, whose concatenation $w(\mathcal{F})$ rearranges S. Equal occurrences are not labeled. The step-one factor conditions are those of [Haw, Definitions 2.3–2.4], with factor and variable indices shifted from $0, 1, \ldots$ to $1, 2, \ldots$.

For integers a, b, define

$$d_\tau(a,b) = \begin{cases} \max(0, a+\tau-b), & a \leq b, \\ \max(0, b+1+\tau-a), & a > b, \end{cases} \qquad \operatorname{dinv}_\tau(w) = \sum_{p<q} d_\tau(w_p, w_q). \tag{4.1}$$

Define the factorization series and its fixed-degree coefficients by

$$\begin{aligned} \mathrm{DS}^{*(\tau)}(S, d; \mathbf{x}) &= \sum_{\substack{\mathcal{F} \text{ dual step-}\tau \\ \operatorname{dinv}_\tau(w(\mathcal{F}))=d}} \prod_{a \geq 1} x_a^{|F_a|}, \\ \mathcal{D}^*_{S,\tau}(\mathbf{x}; q) &= \sum_{d \geq 0} q^d \, \mathrm{DS}^{*(\tau)}(S, d; \mathbf{x}). \end{aligned} \tag{4.2}$$

Each monomial coefficient is finite because its exponents prescribe all factor lengths. There is no condition on the first entry or sign of a factor, and no restriction on S beyond finiteness.

Write a nonempty multiset as $S = c_1^{a_1} \cdots c_r^{a_r}$, with $c_1 < \cdots < c_r$ and $a_i > 0$. Let $G_{S,\tau}$ have vertices $[r]$ and edges

$$\{i,j\} \in E(G_{S,\tau}) \quad \Longleftrightarrow \quad i < j \text{ and } c_j - c_i \leq \tau. \tag{4.3}$$

This is a natural unit interval graph: its defining function is $g(i) = \max\{j : c_j \leq c_i + \tau\}$, which is weakly increasing and satisfies $g(i) \geq i$. Put

$$\begin{aligned} M_\tau(S) &= \sum_{\substack{i<j \\ c_j - c_i \leq \tau}} a_i a_j, \\ \beta_\tau(S) &= \tau \sum_i \binom{a_i}{2} + \sum_{\substack{i<j \\ c_j - c_i \leq \tau}} (\tau - c_j + c_i) a_i a_j. \end{aligned} \tag{4.4}$$

Both quantities are nonnegative integers.

Lemma 4.1. *For every rearrangement w of S,*

$$\operatorname{dinv}_\tau(w) = \beta_\tau(S) + \#\{(p,q) : p < q,\ 0 < w_p - w_q \leq \tau\}. \tag{4.5}$$

Proof. Each equal pair contributes τ. For distinct values with difference $\delta > 0$, the increasing order contributes $\max(0, \tau - \delta)$ and the decreasing order contributes $\max(0, \tau + 1 - \delta)$. Since δ and τ are integers, both are zero when $\delta > \tau$; otherwise they are $\tau - \delta$ and $\tau - \delta + 1$. Summing the order-independent contributions gives $\beta_\tau(S)$, with one extra unit for each counted decreasing pair. □

Theorem 4.2. *Under the external input of Theorem 3.1, for every finite multiset S of integers and all integers $\tau, d \geq 0$, the series $\mathrm{DS}^{*(\tau)}(S, d; \mathbf{x})$ is symmetric and e-positive over $\mathbb{Z}_{\geq 0}$. More precisely, for nonempty S, write $G = G_{S,\tau}$, $\mathbf{a} = (a_1, \ldots, a_r)$, $M = M_\tau(S)$, and $\beta = \beta_\tau(S)$. Then*

$$\begin{aligned} \mathcal{D}^*_{S,\tau}(\mathbf{x}; q) &= q^{\beta+M} X_{G,\mathbf{a}}(\mathbf{x}; q^{-1}) \\ &= \sum_{\lambda \vdash |S|} q^{\beta+M} c_\lambda(G, \mathbf{a}; q^{-1}) e_\lambda(\mathbf{x}), \end{aligned} \tag{4.6}$$

and each displayed elementary coefficient belongs to $\mathbb{Z}_{\geq 0}[q]$. In particular, for $\beta \leq d \leq \beta + M$,

$$[e_\lambda]\, \mathrm{DS}^{*(\tau)}(S, d; \mathbf{x}) = [q^{\beta+M-d}] c_\lambda(G, \mathbf{a}; q) \in \mathbb{Z}_{\geq 0}, \tag{4.7}$$

and $\mathrm{DS}^{*(\tau)}(S, d; \mathbf{x}) = 0$ outside that interval.

Proof. Every dual step-τ factor lists distinct values in increasing order with successive gaps greater than τ. All its pairwise gaps therefore exceed τ, so its value levels form an independent set of G. Assign to vertex i the set $C_i = \{a : c_i \text{ occurs in } F_a\}$. This gives a proper multicoloring with $|C_i| = a_i$. Conversely, from such a multicoloring, form F_a by listing in increasing order all c_i with $a \in C_i$. Independence ensures the required gaps. These constructions are inverse and preserve monomials, including empty factors and repeated values.

For an edge $i < j$, occurrences in factors $a \in C_i$ and $b \in C_j$ form a decreasing pair in (4.5) exactly when $b < a$. Properness excludes $a = b$, so among the $a_i a_j$ pairs on this edge, these are exactly those not counted by asc_G. Every decreasing pair counted there comes from an edge. Consequently, Lemma 4.1 gives

$$\mathrm{dinv}_\tau(w(\mathcal{F})) = \beta + M - \mathrm{asc}_G(C).$$

Summing this monomial-preserving identity proves the first equality in (4.6). By Theorem 3.1, each $c_\lambda(G, \mathbf{a}; q)$ has nonnegative integer coefficients and degree at most M. Reversing its coefficients and multiplying by q^β therefore gives a positive polynomial supported in $[\beta, \beta + M]$. Extracting q^d proves the claims. For $S = \varnothing$, the unique factorization has every factor empty, so $\mathrm{DS}^{*(\tau)}(\varnothing, 0; \mathbf{x}) = 1$ and all other degrees are zero. □

4.1 The original dual Dyck functions

Hawkes's original statistic is

$$\mathrm{di}(w) = \#\{(p, q) : p < q,\ w_p = w_q + 1\};$$

in particular, equal pairs are not counted [Haw, Definition 2.1]. Put $\kappa(S) = \sum_i \binom{a_i}{2}$. At $\tau = 1$, the distinct-value terms in $\beta_1(S)$ vanish and (4.5) becomes

$$\beta_1(S) = \kappa(S), \qquad \mathrm{dinv}_1(w) = \kappa(S) + \mathrm{di}(w). \tag{4.8}$$

Corollary 4.3. *Under the same external input, the original dual Dyck symmetric functions $\mathrm{DS}^*(S, d; \mathbf{x})$ of [Haw, Definition 2.5] are e-positive over $\mathbb{Z}_{\geq 0}$ for every finite multiset S and every $d \geq 0$.*

Proof. The step-one factor conditions agree with Hawkes's, after reindexing the variables. Equation (4.8) gives

$$\mathrm{DS}^*(S, d; \mathbf{x}) = \mathrm{DS}^{*(1)}(S, d + \kappa(S); \mathbf{x}),$$

so Theorem 4.2 applies. The empty multiset is immediate. □

For $\tau = 0$, the value graph has no edges and both $M_0(S)$ and $\beta_0(S)$ vanish, giving $\mathcal{D}^*_{S,0} = \prod_i e_{a_i}$. For a single value repeated m times, the graph has no edges at any step, but the equal-pair shift gives $\mathcal{D}^*_{S,\tau} = q^{\tau\binom{m}{2}} e_m$. For $\tau > 1$, nearby distinct values can also contribute to $\beta_\tau(S)$; the equal-pair term alone would not give the correct grading.

References

[Haw] Graham Hawkes, *Dyck symmetric functions and applications to q,t-Catalan polynomials*, preprint, 2026. arXiv:2605.13003v1. Numbered references above follow this version.

[Hwa] Byung-Hak Hwang, *Chromatic quasisymmetric functions and noncommutative P-symmetric functions*, Transactions of the American Mathematical Society **377** (2024), 2855–2896. doi:10.1090/tran/9096. arXiv:2208.09857v2.

[KL] Young-Hoon Kiem and Donggun Lee, *Birational geometry of generalized Hessenberg varieties and the generalized Shareshian–Wachs conjecture*, Journal of Combinatorial Theory, Series A **206** (2024), 105884. doi:10.1016/j.jcta.2024.105884. arXiv:2208.12282v3, 19 February 2024. Numbered references above follow this arXiv version.

[OAI] OpenAI, *Elementary positivity of chromatic quasisymmetric functions*, OpenAI Math Release preprint, 24 September 2026, 51 pp. OpenAI mathematics repository: manuscript PDF at revision `fd4aeeb`. The external inputs used here are Proposition 2.2, Lemmas 5.1 and 5.7, and Theorem 5.8 of that version.